

Towards Secure Air Traffic Surveillance: A Survey on ADS-B Threats, Existing Solutions, and Future Research

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(Survey Paper)

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ABSTRACT Automatic Dependent Surveillance-Broadcast (ADS-B) has transformed air traffic surveillance systems by improving flight efficiency, reducing operational costs, and enhancing situational awareness. Despite the protocol's advantages, its unencrypted and open nature makes it vulnerable to significant cybersecurity threats, including message injections, message spoofing, jamming, and message modification attacks, which pose serious risks to aviation efficiency and safety. This research survey comprehensively evaluates ADS-B security, analyzing its vulnerabilities, threat landscape, and mitigation methodologies. We systematically categorize state-of-the-art security solutions, including artificial intelligence(AI)-based intrusion detection, cryptographic approaches, signal and time-based solutions, physical-layer security, and blockchain-based ADS-B authentication and integrity frameworks. A comparative analysis of the categorized security solutions highlights their feasibility, challenges, strengths, weaknesses, and real-world implementation trade-offs. Furthermore, this survey paper identifies key challenges to securing ADS-B, such as compatibility with legacy systems, regulatory constraints, and computational limitations. It outlines future research directions to enhance ADS-B resilience. By focusing on the identified security gaps, this research survey proposes a multi-layered and robust security method for ADS-B, ensuring modern air traffic surveillance systems' confidentiality, integrity, availability, and authenticity.

INDEX TERMS ADS-B, security and privacy, air traffic control, aviation security, surveillance, cryptography.

I. INTRODUCTION

The aviation industry makes extensive use of technology. However, technological advancements have also increased the risk of hackers breaking into the systems and exploiting the connections between aviation systems. Worldwide, aviation industry has seen an increase in cyber attacks and information and communications technology (ICT) disruptions in recent years [1]. As the global economy improves, a greater number of people prefer air travel. The Federal Aviation Administration (FAA) forecasts that 1.15 billion passengers will be traveling on commercial aircraft by 2033 [2]. In the foreseeable future, airspace will continue to get ever more crowded as more aircraft are added to meet demand. The

FAA created the NextGen (Next Generation Air Transportation System) project in 2004 to improve flight safety and meet future navigation needs. This project aimed to progressively replace the current land-based point-to-point air traffic control (ATC) system. ADS-B, which is a modern air traffic surveillance technology that enhances and optimizes situational awareness and airspace management, was introduced as a key communication component of the NextGen project by the FAA [1]. ADS-B is an alternative for traditional radar-based tracking (primary and secondary radar surveillance) with satellite-based navigation and real-time aircraft position receiving and broadcasting. ADS-B improves accuracy, coverage, and efficiency, particularly in non-radar environments,

TABLE 1. Cyber Attacks on Aviation Industry in Recent Years

Year	Event Description
2025	Italian airports, including Linate and Malpensa, faced DDoS attacks claimed by the hacker group <i>Noname057(16)</i> , disrupting services and highlighting vulnerabilities in airport cybersecurity [6].
2024	GPS Spoofing Incidents: A surge in GPS spoofing, specifically around conflict zones, caused navigation disruptions for commercial airliners and affected systems reliant on location data and accurate time [7].
2024	Japan Airlines experienced a DDoS attack that disrupted its network systems, leading to delays in over 20 domestic flights [8].
2024	CrowdStrike-Related IT Outage: A faulty update from cybersecurity organization CrowdStrike led to a worldwide IT outage, causing widespread flight cancellations and delays across multiple airlines [9].
2024	Seattle-Tacoma International Airport Cyberattack: A cyberattack disrupted systems at Seattle-Tacoma International Airport, leading to significant delays and baggage handling issues [10].
2023	Ransomware emerged as the leading cyber threat in the aviation sector, accounting for 22% of all malicious incidents. This surge was highlighted by a 600% increase in ransomware occurrences within the aviation supply chain, as reported by Boeing's Chief Security Officer [11].
2021	SITA Data Breach: A cyberattack on airline technology provider SITA affected multiple airlines, compromising passenger information stored in SITA's Passenger Service System servers [12].
2020	EasyJet Data Breach: A sophisticated cyberattack exposed the email addresses and travel information of approximately 9 million customers; 2,208 customers also had their credit card details accessed [12].
2019	Malaysia Airlines Passenger Data Breach: Millions of passport details, phone numbers, and addresses were compromised and uploaded online.
2018	Cathay Pacific Airways Data Breach: Unauthorized access led to the compromise of personal data of 9.4 million passengers, including passport details, birth dates, and credit card information [12].
2018	Bristol Airport (UK) Extortion Attack: The airport's flight information system was compromised, forcing airport staff to use whiteboards and markers for the flight schedule [2].

including remote airspace areas and oceans [3], while reducing the operational costs. ADS-B has become mandatory for aircraft operating in the U.S. and European airspace since 2020 [4].

The importance of ADS-B for situational awareness can be gauged by the recent mid-air collision that took place over the Potomac River on January 29, 2025, near Washington, D.C. The crash involved American Eagle Flight 5342, a regional passenger jet, and a U.S. Army UH-60 Black Hawk helicopter. Tragically, there were no survivors—all 67 people on board both aircraft lost their lives. The investigation later revealed that the Black Hawk's ADS-B system had been turned off during the flight. While military aircraft are allowed to disable ADS-B transmissions for operational security, this particular mission was just a routine training exercise. The decision to fly without ADS-B raised serious concerns about the safety risks this exemption can pose in civilian airspace.

Despite its advantages, ADS-B has certain weaknesses as well. It operates over an open and unencrypted communication channel, making it vulnerable to several cybersecurity threats [5]. ADS-B messages are broadcast in the air for surrounding aircraft and ground-based ATC without authentication or encryption mechanisms. This allows malicious entities to intercept ADS-B messages and modify, inject, or drop critical flight information. Cybersecurity incidents in the aviation industry are rapidly increasing. Table 1 shows the cyberattacks against the aviation industry in recent years. Attackers can exploit ADS-B vulnerabilities for spoofing aircraft positions, creating ghost aircraft, or conducting denial-of-service (DoS) attacks, posing significant risks to ATC operations and passenger safety.

While many security solutions have been proposed for ADS-B [13], [14], [15], no single solution fully protects it from the identified threats. Security techniques such as spread spectrum techniques [16], [17], Public Key Infrastructure (PKI) [18], [19], ML and DL [20], [21], and multilateration-based verification [17], [22] provide some security benefits but also introduce challenges such as computational complexity, high implementation costs, and limited real-world implementation and deployment. Besides, traditional rule-based intrusion detection systems (IDS) often do not detect sophisticated anomalous behavior in ADS-B messages.

Considering the above challenges, there is a need for a robust and efficient security framework capable of detecting and mitigating real-time ADS-B attacks. This research presents a state-of-the-art survey on ADS-B security issues, including a critical evaluation of the existing literature, an assessment of aviation surveillance protocols, a review of ADS-B threats and associated attacks, current challenges, and future research directions. This research aims to contribute towards a more resilient and secure air traffic surveillance system.

A. RESEARCH SIGNIFICANCE

ADS-B is a fundamental part of modern air traffic management, providing reduced operational costs, enhanced situational awareness, and increased flight efficiency. However, the protocol's open and unencrypted nature introduced several critical security threats, making it a leading target for attackers. Addressing these threats is necessary to ensure the safety, reliability, and resilience of global airspace operations. The significance of this research is highlighted in the following points:

- ADS-B has been increasingly targeted by several attacks, including jamming, spoofing, message injection, and denial-of-service, which pose severe security risks to air traffic operations and flight safety.
- Traditional IDS and encryption-based methods offer some solutions but fail to provide a comprehensive defense against growing attacks.
- Considering the complexity of the protocol security challenges, a multi-layered security framework integrating anomaly detection using AI, cryptographic authentication, and hybrid security mechanisms is crucial for strengthening ADS-B resilience.
- The FAA, International Civil Aviation Organisation (ICAO), and EUROCONTROL have recognised the threats associated with ADS-B. However, there is still no standardised, internationally accepted cybersecurity solution to protect ADS-B communication from these threats. Research in this area is crucial for developing regulatory frameworks and informing future policy.

B. RESEARCH QUESTIONS

The following are the research questions for the proposed research paper:

- *Q1*: What are the capabilities, limitations, and security implications of current aviation surveillance systems?
- *Q2*: What are the key threats and attack vectors against ADS-B systems, and how can a comprehensive threat model inform the design of effective countermeasures?
- *Q3*: What are the current state-of-the-art security solutions for ADS-B security, and what are their strengths and weaknesses in addressing identified threats?
- *Q4*: What are the potential challenges of implementing ADS-B security solutions, and how can they be addressed?
- *Q5*: How does the operational and regulatory framework of ADS-B influence its security posture?
- *Q6*: What are the key open challenges in ADS-B security, and what future research directions need to be explored to improve the resilience and trustworthiness of ADS-B?

C. RESEARCH CONTRIBUTIONS

The following are the main contributions of our research:

- We investigate the state-of-the-art aviation surveillance systems, highlighting their capabilities, limitations, and security implications, and examine real-world avionics security incidents and attack scenarios, offering insights for the development of more robust detection, prevention, and response strategies.
- We provide an in-depth technical analysis of ADS-B, covering its operational framework, regulatory landscape, security vulnerabilities, and publicly available datasets of ADS-B traffic.
- We provide a comprehensive ADS-B threat model and propose a novel taxonomy based on an integrated view of ADS-B security requirements, aiming to provide a

holistic understanding of current solutions, verification techniques, evaluation parameters, threats, and security goals.

- We conduct a critical review of state-of-the-art security solutions for improving ADS-B security, including ML-based intrusion detection systems, cryptographic techniques, and physical-layer security solutions. Additionally, we perform a comparative analysis of the existing solutions, evaluating their strengths, weaknesses, feasibility, and performance trade-offs.
- We designed a novel taxonomy for ADS-B security, identified open security challenges, and outlined future research directions to address existing gaps, paving the way for more resilient ADS-B security solutions.

D. PAPER STRUCTURE

The rest of the paper is structured as follows: Section II presents the research methodology, including the research paper selection process, the inclusion and exclusion criteria, and the keyword search strategy. Section III discusses existing survey papers related to ADS-B security. Section VI comprehensively reviews aviation surveillance protocols. Section IV presents the technical background of the ADS-B, including system architecture, communication and message format, whereas Section V presents the ADS-B threat model, associated vulnerabilities, existing open sources databases, and regulatory frameworks and presents a comprehensive categorisation of existing solutions. Section VIII presents a comprehensive integrated view of ADS-B security in the form of a taxonomy. Section IX presents the challenges, future research directions, and cross-cutting challenges. Finally, Section X concludes this survey paper.

II. RESEARCH METHODOLOGY

The research methodology section outlines a systematic literature review carried out to identify and examine research on ADS-B security. Using predefined inclusion and exclusion criteria and a structured keyword search across major digital libraries, we conducted staged screening (title/abstract), removed duplicates, assessed full texts, and prioritised novel detection and mitigation frameworks. The paper selection process is summarised in an adapted PRISMA-style flow (Fig. 1).

A. INCLUSION AND EXCLUSION CRITERIA

In the research survey, we included peer-reviewed papers and conference articles (from high-impact journals and reputable conferences) published between 2014 and 2024 that specifically address ADS-B security (confidentiality, integrity, availability, and related countermeasures). Items that were duplicates or redundant works from the same authors, publications without an ADS-B security focus, editorials, patents, and non-peer-reviewed sources were excluded.

B. KEYWORD SEARCH STRATEGY

A systematic, keyword-based search was conducted using Boolean operators across major digital libraries, including

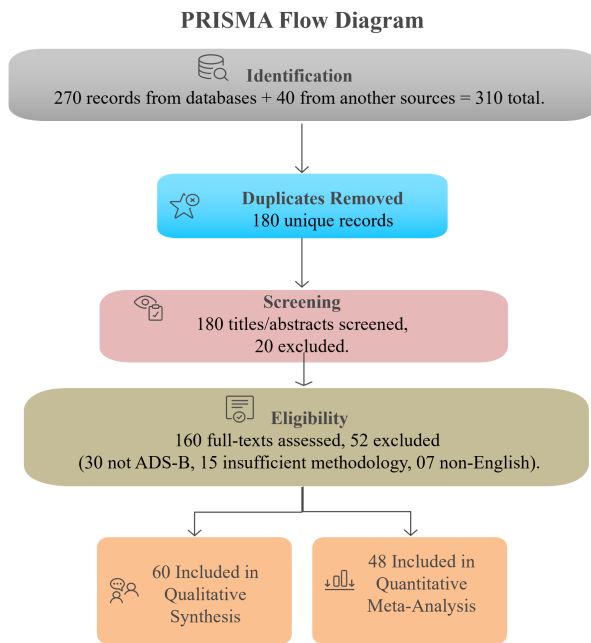


FIGURE 1. PRISMA Flow Diagram For the Proposed Survey Paper.

Springer, IEEE Xplore, ACM Digital Library, ScienceDirect, and Google Scholar. Typical search strings included:

- “ADS-B security” OR “Automatic Dependent Surveillance-Broadcast security”
- “ADS-B spoofing mitigation” OR “cryptographic techniques in ADS-B”
- “ADS-B jamming attacks” OR “ADS-B message integrity”
- “aviation cybersecurity” OR “ADS-B vulnerabilities” OR “ADS-B attacks”
- “ADS-B intrusion detection” OR “machine learning for ADS-B security”

C. RESEARCH PAPER SELECTION PROCESS

The selection followed a staged screening process: (1) initial screening of titles and abstracts to discard obviously irrelevant records, (2) removal of duplicates, (3) full-text assessment for relevance to ADS-B security, and (4) prioritisation of novel frameworks, followed by categorisation by domain/technology and a final comparative analysis of short-listed works (feasibility, limitations, trade-offs). This process is illustrated in the Fig. 1 (PRISMA Flow Diagram For the Proposed Survey Paper).

III. SURVEYS ON ADS-B

Several survey papers have explored ADS-B vulnerabilities, threats, and possible mitigation strategies and techniques. The previous surveys presented a broad spectrum of attack scenarios, countermeasures, and advancements in ML, DL, and cryptographic techniques, providing the foundation for understanding the current state of ADS-B security. This section summarizes the existing surveys and compares our work’s contributions with those of others.

Authors in [23] systematically review the security vulnerabilities and provide an overview of existing attacks and countermeasures of ADS-B, whereas in [2], authors categorise ADS-B vulnerabilities based on attack intentions and security requirements. The survey paper evaluates existing security solutions, such as secure location verification and broadcast authentication, analysing their pros and cons. In [16], the authors analysed the vulnerabilities of the ADS-B system and categorised identified attacks based on the security requirements and protocol layers. It evaluates state-of-the-art countermeasures, highlighting their pros, cons, and implementation challenges. In [4], the authors comprehensively review ADS-B, detailing its operational principles, infrastructure, and applications. It highlights the ADS-B’s vulnerabilities, particularly the lack of encryption and authentication, whereas [24] reviews the vulnerabilities affecting ADS-B systems, categorising them into message modification, injection, eavesdropping, jamming, and deletion. The authors examine mitigation strategies such as timestamp authentication, multilateration, encryption methods, and multichannel receivers. They also highlight gaps in the current security mechanisms, focusing on the limitations of existing countermeasures and their implementation complexities. In [25], the authors review the security concerns of the ADS-B protocol, highlighting its susceptibility to jamming, spoofing, and injection attacks. The authors explore cryptographic and non-cryptographic security measures, such as multilateration, public key cryptography, and retroactive key publication.

Considering the contributions and limitations of the previous research work, the proposed survey provided a comprehensive ADS-B threat model, systematically identifying potential threats and attacks based on security requirements. Based on the threat model, we provided an in-depth technical analysis of ADS-B, covering its operational framework, regulatory landscape, security vulnerabilities, and publicly available datasets of ADS-B traffic. The survey also critically reviewed state-of-the-art security solutions for ADS-B, including ML-based intrusion detection systems, cryptographic techniques, and physical-layer security solutions. Table 2 compares our work with existing surveys on ADS-B security.

IV. ADS-B TECHNOLOGY

A. REGULATORY FRAMEWORK

ADS-B is an important part of modern ATMs, designed to enhance situational awareness and surveillance capabilities. Key aviation regulatory bodies govern its implementation and operation [26]. The ICAO manages it globally, the FAA in the United States, and EUROCONTROL in Europe.

1) FAA

The FAA has mandated ADS-B Out equipment in the United States for aircraft operating in designated airspace since January 1, 2020 [26]. This requirement is detailed in Title 14 of the Code of Federal Regulations, specifically:

TABLE 2. Comparison of Our Work With Existing Surveys on ADS-B Security

Domains	Our Survey	[5]	[2]	[24]	[16]	[25]
Compare with existing surveys	✓	✓	✓			
Aviation Surveillance Protocols	✓					
Technical Overview	✓	✓	✓	✓	✓	✓
Threat Model	✓		✓			
Security Vulnerabilities	✓	✓	✓	✓	✓	✓
Real-world Datasets	✓					
Regulatory Frameworks	✓	✓			✓	
Comprehensive solution taxonomy	✓	✓	✓	✓	✓	✓
Comparative Analysis of Security Solutions	✓	✓		✓	✓	✓
Attacks Case Study	✓					
Open Challenges	✓		✓			
Research Opportunities	✓	✓	✓			✓

- *14 CFR § 91.225*: This title states the airspace where ADS-B Out equipment is required.
- *14 CFR § 91.227*: This title states the functioning requirements for ADS-B Out equipment, including the standards of integrity and accuracy.

The above regulations have enhanced the safety and efficiency of the national airspace and ensured that aircraft transmit precise position information.

2) ICAO

ICAO is a dedicated and specialised agency whose role is to formulate and implement international standards and recommended practices for the efficiency and safety of the aviation industry [27]. ICAO encapsulates the formulated standards for ADS-B, including [27]:

- *Doc 9871*: This standard offers technical support for Mode S and Extended Squitter and the implementation guidelines for ADS-B.
- *Annexe 10 to the Convention on International Civil Aviation*: This standard offers technical characteristics for aviation communication systems, including ADS-B.

These standards guide member states in harmonising ADS-B implementation, ensuring global standardisation and interoperability.

3) EUROCONTROL

EUROCONTROL is responsible for the European Union's Aviation Safety [28]. It has formulated mandates for ADS-B equipment to enhance safety and improve the efficiency of air traffic surveillance. The primary standard is:

- *Commission Implementing Regulation (EU) No 1207/2011*: This regulation lays down the requirements for air traffic surveillance performance and interoperability in the Single European Sky [28].

These standard rules and regulations mandate that aircraft operating under instrument flight rules be equipped with ADS-B Out by the specified deadlines, with certain exemptions for state aircraft.

B. ADS-B SYSTEM ARCHITECTURE

ADS-B is an important component of the next-generation aircraft surveillance technology project designed to enhance the efficiency of ATM by automatically broadcasting aircraft information, including identity, position, velocity, and intent information [29]. Unlike traditional radar-based systems (primary and secondary radar), ADS-B depends on aircraft transponders and satellite navigation technology to provide precise, accurate, real-time tracking [30]. Regulatory bodies such as the FAA and EUROCONTROL have mandated ADS-B since 2020. ADS-B is divided into two modes from the perspective of operations [31]:

- *ADS-B Out*: The aircraft generates precise location data from the navigation system, processes this data by adding aircraft identity information, and broadcasts it to nearby aircraft and ground stations.
- *ADS-B In*: The aircraft receives data from other ADS-B Out-equipped aircraft and ground stations for real-time surveillance and situational awareness.

The advantages of the ADS-B include improved aircraft separation, increased situational awareness, cost efficiency, and enhanced safety compared to traditional radar systems [32]. However, the protocol's open broadcast introduces several critical security vulnerabilities, necessitating robust security measures to protect against cyber threats.

C. ADS-B COMMUNICATION CHANNELS

ADS-B communication depends on radio frequency (RF) transmission. Aircraft with ADS-B Out capability continuously broadcast their positional information (GPS-derived position) twice a second and identity information every 5 seconds using [33]:

- *Universal Access Transceiver (UAT)*: Operates on 978 MHz and is used by general aviation.
- *1090 MHz Extended Squitter (1090ES)*: Primarily used by high-performance and commercial aircraft.

The ADS-B signal transmission includes aircraft identification (ICAO address, call sign, etc.), position information

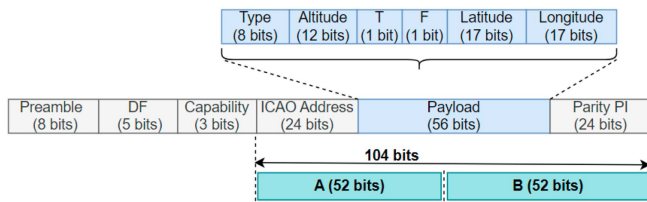


FIGURE 2. ADS-B Protocol Message Format.

(velocity and altitude data), and emergency status and intent messages. The broadcast messages are received by ATC facilities, ground stations, and nearby aircraft, enabling real-time air traffic visualisation for improved situational awareness [34]. Although it has operational advantages, the lack of authentication and encryption methods makes ADS-B highly vulnerable to several cyber threats, including message injection, spoofing, and jamming attacks.

D. ADS-B MESSAGE FORMAT

The ADS-B message is a standardised 112-bit long, including a 5-bit downlink Format (DF) and a 3-bit capability Code (CA), which identify the message type [35]. This is followed by a 24-bit ICAO aircraft address assigned to each aircraft, 56-bit space for data containing aircraft position, velocity, and status information, and a 24-bit parity check (PI), a simple integrity check to detect transmission errors [35] as illustrated in Fig. 2 [36].

V. ADS-B SECURITY THREATS AND ATTACKS

The inherent vulnerability of the ADS-B protocol is due to its lack of encryption, open broadcast nature, and authentication mechanisms. Unlike traditional surveillance protocols, which depend on ground-based verification or transponder interrogation, ADS-B messages are continuously broadcast by aircraft without any form of handshake or challenge-response. Any attacker with low-cost software-defined radio (SDR) equipment can easily intercept, decode, and even inject false ADS-B messages into the system. The absence of encryption techniques allows attackers to access sensitive flight information. At the same time, the lack of authentication makes it trivial to create ghost targets, spoof aircraft positions, or disrupt situational awareness.

A. ADS-B THREAD MODEL

There are two main communication channels in ADS-B operations:

- *Air Communication:* The aircraft equipped with ADS-B request location from satellites and receive responses, enabling them to calculate their precise positions. Aircraft broadcast ADS-B Out messages, which contain real-time flight and location information. The surrounding aircraft and ground stations receive the broadcast message for safety and situational awareness.
- *Ground Communication:* Ground station receivers receive ADS-B messages from aircraft and forward them

to ATC. The ATC monitors and manages air traffic based on received ADS-B data for situational awareness and airspace safety.

Both air-based and ground-based attacks pose serious threats to aviation security. Attackers may introduce non-existent aircraft, causing confusion and misleading pilots and ATC about actual aircraft positions. They may disrupt aircraft coordination by injecting false data and prevent ATC from detecting real aircraft using jamming attacks.

Fig. 3 describes the communication architecture of ADS-B and the proposed threats model (spoofing, jamming, message injection, Eavesdropping, and message modification) across air and ground segments. In this threat model, attackers are classified into two categories according to their setting: ground-based attackers (If attacker-modified data enters at this point, ATC's situational picture can be manipulated—leading to misrouting, ghost aircraft, or traffic conflicts) and air-based attackers (An attacker aircraft can inject fake ADS-B signals, pretending to be a legitimate aircraft or altering information). The threat model captures communication between satellites, aircraft, ground stations, and ATC, highlighting key vulnerabilities, threats, and attack vectors. ADS-B security threats arise mainly from its unsecured transmission [37].

In this survey paper, we aim to provide a clearer understanding of the diverse threat landscape affecting ADS-B systems by introducing a structured threat taxonomy based on both the OSI model layers and ADS-B security requirements (CIA+A), as illustrated in Fig. 4 and Table 3. This dual taxonomy enables a more systematic classification of attacks, helping to identify which layer or property the system targets. By organising threats this way, researchers and system architects can better prioritise defences based on the exposure layer and the type of security violation involved.

The most considerable security threats include:

- *Eavesdropping:* ADS-B transmits unencrypted flight data, allowing attackers to track aircraft in real-time and potentially compromising privacy, particularly for VIP and military flights. Eavesdropped data can also be used later for spoofing.
- *Message Deletion:* Attackers corrupt or jam the ADS-B protocol signals, making aircraft invisible to ATC and nearby aircraft.
- *Message Modification:* This includes the unauthorised modification of legitimate ADS-B messages and altering aircraft position or identification data.
- *Message Injection:* Attackers create ghost aircraft, inject false aircraft data or modify an aircraft's position information, such as velocity.
- *Message Jamming:* The ADS-B jamming threat involves intentionally transmitting radio frequency signals to interfere with the reception of ADS-B messages. By overwhelming the ADS-B frequency band, attackers can block the reception of critical surveillance data.
- *Message Flooding:* The ADS-B flooding threat involves transmitting a high volume of ADS-B messages, either modified or genuine, to disturb the system.

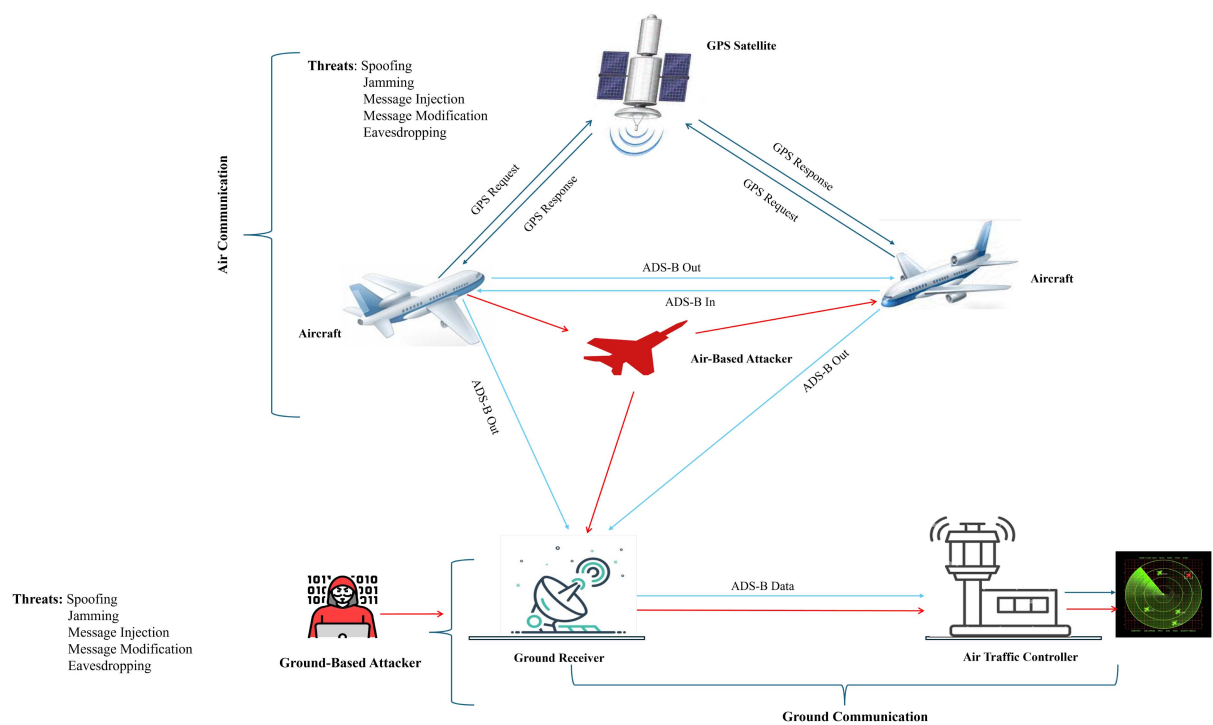


FIGURE 3. ADS-B Threat Model.

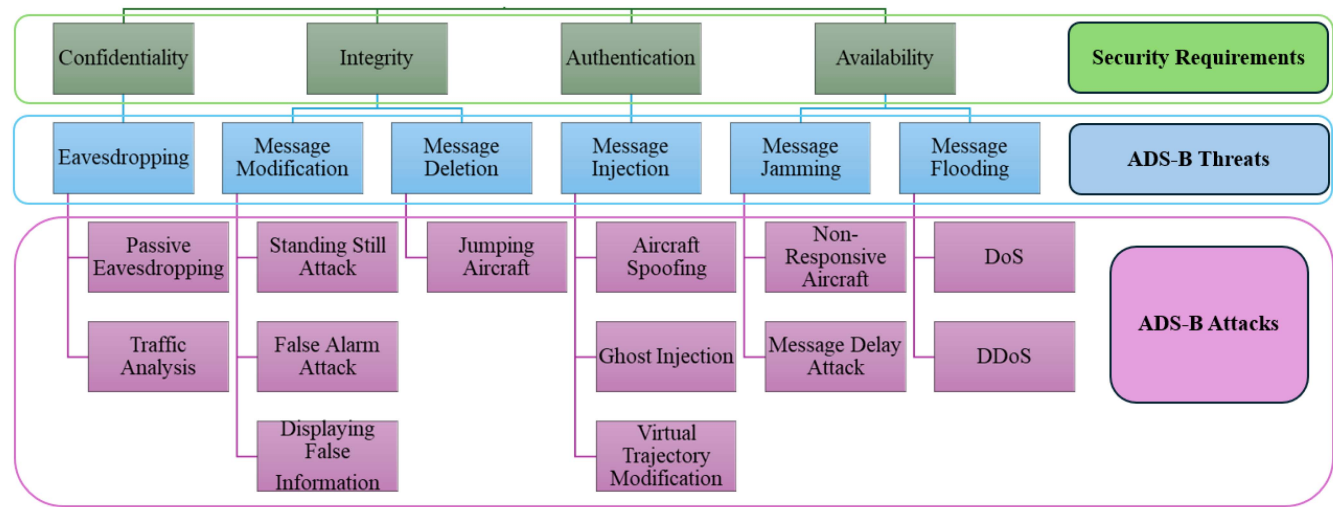


FIGURE 4. ADS-B Security Requirements, Threats and Attacks.

Fig. 4 illustrates how high-level security goals cascade into threats, which then lead to specific attack manifestations. The top row of the figure presents the security requirements for ADS-B from the perspective of CIA+A. These are the properties that the surveillance system must uphold. The middle row categorises the threat types that directly violate these requirements: eavesdropping attacks compromise confidentiality; message modification and deletion undermine integrity (and often authentication); message injection targets authentication and integrity through the injection of forged frames; and message jamming or flooding

attacks aim to disrupt availability by denying service. The lower section displays attack manifestations — concrete ways an adversary carries out those threats within the ADS-B ecosystem.

Table 3 shows layered threat mapping for ADS-B. Attacks are linked to the OSI layers where they occur: Physical, Data Link, and Network/Backend. Message deletion typically occurs due to CRC/bit errors caused by RF interference; message flooding can occur at either the RF level (air-link) or the network level (backend). This mapping clarifies where mitigations such as sensor fusion, multilateration, RF

TABLE 3. ADS-B Sub-Attacks Mapped to OSI Layers, Attack Mechanism and Impact on CIA+A

OSI Layer	Sub-attack	Attack Mechanism	Impact on CIA+A
Physical	Eavesdropping	Passive reception of clear-text ADS-B frames	Confidentiality
	Jamming and RF Flooding	Overwhelming RF channel or raising noise floor	Availability
Physical and supports Data-Link	Message Deletion (RF loss or CRC errors)	CRC/bit errors or RF loss causing frame drop	Availability and Integrity
	RF Replay and Reactive Spoofing	Re-transmit captured waveforms or reactively mimic signals	Availability and Integrity
Data-Link	Message Injection (forged Mode-S, 1090ES and UAT)	Crafting valid-looking Mode-S, 1090ES, and UAT frames	Integrity and Authentication
	Spoofing (Ghost aircraft and identity impersonation)	Forging ICAO, position, and velocity fields in squitters	Integrity and Authentication
	Message Modification and Bit-flipping	Altering fields or flipping bits to change content or CRC	Integrity
	Standing-Still, Jumping, and Virtual trajectory modification	Manipulating successive broadcasts to show unrealistic motion	Integrity and Authentication
	Message Delay (airlink-level latency and buffering)	Hold or late replay of valid frames to corrupt timing	Integrity and Availability
Network, Transport, and	Backend message injection, and modification	Altering ADS-B records after ground ingest (DB or middleware tampering)	Integrity
Application (ground/backend)	Backend DDoS, and Server DoS (ATC or aggregator)	Overwhelming ground servers or distribution services	Availability
	Message flooding (network and backhaul)	Excessive packets or malformed feeds over IP links to ground infra	Availability
	Non-responsive and missing track (backend)	Processing faults or targeted attacks preventing track forwarding	Availability
	Traffic analysis & metadata abuse	Aggregation/correlation of stored feeds or logs to profile movements	Confidentiality

TABLE 4. Comparison of ADS-B Datasets

Feature	FlightAware [38]	FlightRadar24 [39]	OpenSky Network [40]	ADS-B Exchange [41]	FAA Database [42]	OpenScope Network [43]
Type	Commercial	Commercial	Open access	Open source	US Government	Open access
Data Types	Real-time and historical	Real-time and historical	Raw	Raw	Historical	Simulated
Coverage	Global	Global	Global	Global	U.S	Global
Paid Plans	API	API	Non-commercial	Commercial use	N/A	No
Free Access	Basic features	Basic features with delay	Research and education	Full raw data for personal use	Public access	Yes
API Access	Yes	Yes	Yes	Yes	No	Yes

fingerprinting, GNSS anti-spoofing, and backend hardening should be implemented.

Physical Layer: The RF link (1090 MHz ES and 978 MHz UAT) preamble and pulse waveforms, as well as timing, are where the raw radio energy is transmitted and received. Attacks and effects that act on the waveform (jamming, eavesdropping, pulse/noise interference, bit errors) occur at this layer. GNSS (GPS) signals that supply position and time are also physical-layer dependencies.

Data-Link Layer: The ADS-B message fields (ICAO ID, position, velocity, and status), as well as the parity/CRC mechanisms, are the Data-Link Layer's responsibilities. Crafting, modifying, replaying or forging ADS-B message fields (spoofing, injection, bit-flip corruption, CRC failures (message deletion)) are Data-Link attacks.

Network, Transport, and Application Layers (ground/backend): Once ground receivers collect ADS-B messages, they are forwarded over IP networks, stored, fused and presented by ATC/backend applications. These higher layers are not part of the ADS-B airlink but are part of the overall system

architecture and can be targeted by network/back-end threats (DDoS, backend message tampering, database compromises, latency/throughput faults).

B. ADS-B DATASETS

There are different ways to access the ADS-B data, including FlightAware, FlightRadar24, OpenSky Network, ADS-B Exchange, FAA Database, and OpenScope Network. Table 4 presents the comparison of these sources. While most of them provide real-world and authentic ADS-B messages, OpenScope is the only environment that provides both authentic and malicious messages. OpenScope is an open-access simulation program. Several researchers have worked on the OpenScope and contributed new attack types to the environment. The FAA database is reliable, but it is limited to the US airspace. OpenSky provides historical and real-time data for education and research purposes. On the other hand, ADS-B Exchange, FlightAware, and FlightRadar24 offer paid subscriptions for real-world datasets.

VI. AVIATION SURVEILLANCE TECHNOLOGIES

The technologies used in aviation surveillance have evolved, transitioning from manual systems to sophisticated digital technologies designed to ensure safety, efficiency, and security in air traffic management.

A. PRIMARY SURVEILLANCE RADAR (PSR)

PSR is one of the foundational technologies used in aviation surveillance. It operates as a non-cooperative radar system that does not rely on onboard equipment, making it critical for detecting unidentified or non-compliant aircraft [44]. PSR functions by transmitting high-frequency radio waves in a specific direction. These waves reflect off any object, such as an aircraft, and return to the radar antenna. The system calculates the object's range by measuring the time the reflected signal returns. One of the primary advantages of PSR is its ability to detect non-cooperative targets. Unlike secondary radar systems, PSR does not require aircraft to carry transponders or other identification equipment, making it ideal for tracking unknown, unauthorized, or military aircraft. Despite its advantages, PSR has several limitations. PSR only provides information about an aircraft's position and direction, but cannot accurately determine its altitude or identity. PSR signals are often affected by reflections from non-aircraft objects, such as terrain, buildings, and weather phenomena, leading to potential false alarms. PSR is a backup to secondary radar systems for air traffic control, particularly in areas where transponder-based systems may not cover all airspace or transponders fail. PSR also locates distressed aircraft when onboard systems are non-functional [45].

B. SECONDARY SURVEILLANCE RADAR (SSR)

SSR represents a significant advancement over PSR by enabling two-way communication between ground radar stations and aircraft transponders. It is widely used in civil and military aviation and forms the backbone of modern air traffic control (ATC) systems worldwide [46]. SSR operates by transmitting interrogation signals from ground stations to aircraft equipped with transponders. The transponders receive the interrogation and respond with encoded data that includes Identification (Mode A), Altitude (Mode C) and Data Link (Mode S). SSR relies on the time delay between the transmission and reception of signals to calculate the aircraft's position. Unlike PSR, SSR provides richer information with higher accuracy. SSR cannot detect aircraft that lack functioning transponders, making them vulnerable to non-cooperative targets [47]. SSR is critical for tracking and managing commercial flights within controlled airspace. Mode S is integral to traffic alert and collision avoidance systems (TCAS), providing situational awareness and conflict resolution.

C. MODE S PROTOCOL

Mode S, which stands for Selective, is a modern enhancement of the SSR protocol designed to address the limitations of older SSR Modes A and C. Mode S provides more efficient communication between ground-based radars and aircraft

transponders by enabling selective interrogation, reducing signal clutter, and supporting advanced functionalities such as TCAS and ADS-B. Mode S works by assigning each aircraft a unique 24-bit address, known as the ICAO address. This allows ground radar systems to interrogate specific aircraft selectively rather than broadcasting to all aircraft in range. Mode S provides richer data than Modes A and C, including identification, altitude, and flight status. It also extends adequate radar coverage by supporting high-power interrogation and response signals [48]. Mode S requires ground-based radar systems capable of handling selective interrogation.

D. WIDE AREA MULTILATERATION (WAM)

WAM is a surveillance technology used in ATMs to track aircraft by measuring the time difference of arrival (TDoA) signals emitted from aircraft transponders [49]. WAM enhances situational awareness by providing precise aircraft position data, complementing traditional radar systems and newer technologies like ADS-B. WAM relies on transponder signals emitted by aircraft. Multiple ground-based sensors positioned in a network receive these signals. By analyzing the time difference at which each sensor receives the signal, WAM calculates the aircraft's position [50]. WAM extends surveillance to areas with limited radar coverage, such as mountainous regions or oceanic airspace. WAM requires a dense network of ground-based sensors, which may be challenging to deploy in certain regions.

E. TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)

TCAS is an onboard system that enhances flight safety by preventing mid-air collisions, provides pilots with real-time alerts about nearby aircraft, and offers recommended actions to avoid potential conflicts. It operates independently of ground-based ATC, using signals from aircraft transponders to determine proximity, altitude, and collision risk [51]. It is widely adopted in commercial and military aviation, forming a critical layer of safety in congested airspace. TCAS works by interrogating transponders on nearby aircraft using SSR frequencies. TCAS reduces the risk of mid-air collisions, even in areas with limited ATC coverage, but it cannot detect non-cooperative aircraft or those without functioning transponders [52].

F. SPACE-BASED ADS-B

Space-based ADS-B is a revolutionary extension of the ADS-B technology, providing real-time global satellite surveillance of aircraft. Unlike terrestrial ADS-B, which relies on ground-based receivers, space-based ADS-B leverages satellite networks to cover remote and oceanic regions where ground stations are impractical [53]. Space-based ADS-B functions by capturing ADS-B Out signals broadcast by aircraft transponders. These signals are relayed to satellites equipped with ADS-B receivers, transmitting the data to ground control centres. It provides real-time surveillance in remote and

oceanic airspace, improving collision avoidance and search-and-rescue operations [54]. The limitations include increased traffic in busy airspace, which leads to overlapping ADS-B signals and complicates satellite reception.

G. AUTOMATIC DEPENDENT SURVEILLANCE-BROADCAST

ADS-B is a modern surveillance technology that enables precise and real-time tracking of aircraft. ADS-B is part of the NextGen air traffic management project, which aims to replace traditional radar-based systems. ADS-B enhances situational awareness and operational efficiency in controlled and uncontrolled airspace by automatically broadcasting its position data twice a second [16]. ADS-B uses onboard avionics to determine the aircraft's position via the GNSS. This data, combined with other flight information, is broadcast periodically without requiring interrogation from ground radar.

Key Finding: Modern air traffic surveillance combines cooperative and non-cooperative sensors to generate the operational picture used by controllers and flight crews. PSR detects aircraft by reflecting radio energy off the airframe, providing non-cooperative, tamper-resistant baseline tracks independent of onboard equipment. In contrast, SSR/Mode S improves PSR by interrogating transponders for identity and altitude information, but it depends on a functioning transponder. WAM/MLAT calculates time-of-arrival differences from transponders to produce precise positions in terminal areas. At the same time, airborne systems like TCAS offer tactical, short-range collision avoidance alerts to pilots. ADS-B differs from these technologies by continuously broadcasting GNSS-derived position, velocity, and intent using ADS-B Out, and when equipped with ADS-B In, providing cockpit traffic awareness. This capability has been transformative for trajectory-based operations, reducing separation minima and enhancing surveillance in regions with limited coverage (including satellite relay for oceanic airspace).

Along with these advantages, ADS-B's dependence on broadcast and GNSS introduces specific limitations, making it a complementary system rather than a complete replacement for legacy sensors. Because ADS-B messages are transmitted in clear text and rely on GNSS for position and timing, they are vulnerable to passive tracking, spoofing, and jamming; in contrast, PSR remains operational during transponder or GNSS outages but is expensive and has coverage gaps. Multilateration and SSR can serve as additional verification layers, confirming ADS-B positions where sensor density permits, while TCAS provides an independent airborne safety net for resolving short-range conflicts. In practice, operational resilience is achieved through sensor fusion and layered redundancy: combining feeds from PSR, SSR, WAM, and ADS-B reduces false alarms and helps identify anomalous ADS-B behaviour. However, this fusion involves trade-offs in cost, infrastructure, latency, and regulatory complexity that any proposed security or deployment framework must address.

Key Takeaways: ADS-B provides transformative cooperative surveillance but must be used alongside

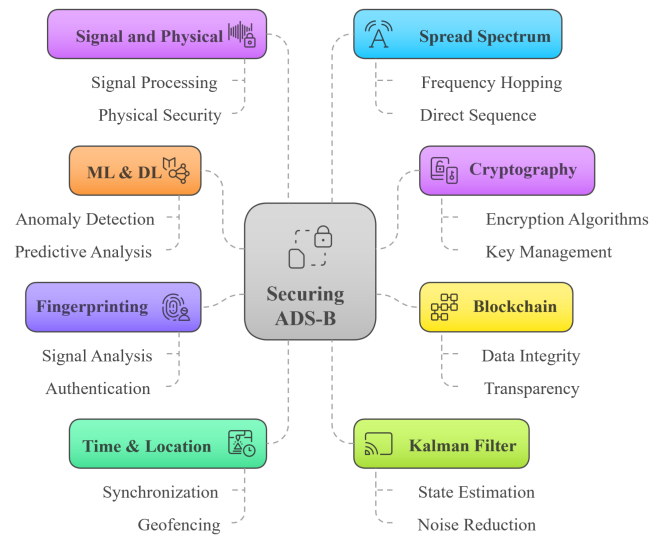


FIGURE 5. Existing security techniques for ADS-B security.

PSR/SSR/WAM/TCAS in a layered, fused architecture to reduce vulnerabilities related to GNSS and broadcast systems.

VII. EXISTING ADS-B SECURITY SOLUTIONS

In the existing literature, aviation engineers, scientists, and academic researchers have proposed security frameworks [18], [31], [55], [56] and solutions to mitigate ADS-B threats using ML-based anomaly detection, multilateration (MLAT), and cryptographic authentication [57]. However, no fully implemented security solution has been developed, leaving ADS-B vulnerable to cyberattacks. This section summarizes the merits and demerits of existing solutions. Many existing security measures require modifications to the ADS-B infrastructure, which makes them unfeasible [16]. Additionally, issues such as software and hardware compatibility and the added burden on the 1090 MHz channel are often overlooked, further reducing the feasibility of these solutions [2].

We categorize current security solutions into different domains. Fig. 5 visually represents this categorization. The following subsections will explain each category, drawing on relevant literature to clarify the concepts and approaches within each domain.

A. MACHINE LEARNING (ML) AND DEEP LEARNING(DL)-BASED SOLUTIONS

In the realm of cybersecurity, AI has emerged as a powerful tool, particularly in anomaly detection, intrusion detection, and attack classification. The adaptability of ML algorithms makes them especially well-suited to address the dynamic and evolving challenges of cybersecurity, offering practical solutions for identifying and mitigating cyber threats [58]. The classification of ADS-B attacks is of paramount significance, as it not only empowers ATC and security experts to enhance existing security measures but also aids in identifying the

origin of attacks. Numerous researchers have developed various solutions to detect anomalies within ADS-B messages. However, the literature on ML-based automatic ADS-B attack classification remains limited. Most existing research primarily categorizes ADS-B messages into two basic categories: malicious and non-malicious. This limitation highlights the need for further research and development in this critical area of aviation security.

In [59], the authors presented an ML-based approach for classifying and detecting jamming attacks, highlighting the importance of ML in this research area. Several supervised ML models, including decision trees, support vector machines, artificial neural networks, and k-nearest neighbours are implemented on a simulated dataset, and their accuracy is compared with the highest accuracy of an artificial neural network (81%), whereas in [59], the authors created a dataset using an ADS-B transmitter. The authors did not share the dataset for future research work.

In [60], the authors proposed an ML-based multi-class classification framework to enhance the security of ADS-B by employing random forests, support vector machines, and decision tree models for classifying ghost aircraft and replay attacks. The authors found that decision tree performance was the best with 92% accuracy among the implemented models. In [61], the authors proposed a spoofing attack detection method based on a two-step DNN for aircraft and message classification, whereas in [62], the authors proposed a classification framework based on the k-nearest neighbor, logistic regression, and naïve Bayes models. The authors generated a dataset using OpenSky API with different attacks: false squawk, false information, name jumping, and false heading attack. The k-nearest neighbour model outperformed other models with an accuracy of 99.57%. On the other hand, a real-time intrusion detection system for ADS-B was proposed by [63] using a support vector machine model with 80% accuracy. The authors trained the model on a dataset generated with the help of OpenSky.

Authors in [64] proposed a long short-term memory (LSTM) model based on DL architecture for detecting malicious ADS-B messages. The model was trained on the OpenSky dataset with 91.59% testing accuracy. From the perspective of ADS-B messages, a spoofing attack detection technique based on LSTM was proposed by [65]. In this research paper, the author focused on three different ADS-B attacks: injection, jamming, and modification. The author implemented an LSTM network model with 93% accuracy. To detect ADS-B spoofed data, the authors in [66] used adversarial learning techniques with 98.87% accuracy. The authors also used the conventional neural network (CNN) technique for aircraft classification based on the ADS-B signals and I/Q samples with 99% accuracy. To detect ADS-B message modification attacks, the authors in [67] proposed an LSTM-based DL technique with 98% accuracy. The authors generated a small, simulated dataset comprising malicious and non-malicious messages using a false data generator tool. The authors in [68] proposed an anomaly detection model

for ADS-B data that had 97% accuracy using a Generative Adversarial Network. The dataset was generated from the Flightradar24 API.

In [69], the authors review machine and deep learning methods for securing ADS-B air traffic control systems, develop a detailed attack model, and propose an IDS using TabNet, NODE, and DeepGBM. Among these, DeepGBM achieved the best performance (98% accuracy), highlighting its effectiveness in detecting ADS-B intrusions and guiding future security frameworks. A Deep Residual Network (DRN) model is proposed by [70] to improve recognition of ADS-B radiated source signals, particularly under low signal-to-noise ratio (SNR) conditions. By leveraging a specialised residual structure, the network learns deeper features without degradation, while a phase-solving preprocessing method is employed to test robustness against fake messages. Experiments show the DRN achieves 95.64% accuracy at 3 dB SNR (24% higher than CNN) and maintains 85.68% accuracy when identifying fake messages, demonstrating strong robustness and reliability. In [71], the authors examine the effect of missing ADS-B/GPS data on flight safety, using data from six aircraft in the OpenSky Network. Various regression models (Bayesian Ridge, Random Forest, AdaBoost, Extra Trees, and k-NN) were tested with random missing rates ranging from 10% to 30%. Results indicate that k-NN consistently outperformed the other models, achieving the lowest MAE values across key parameters (e.g., latitude: 0.0845, longitude: 0.203, altitude: 25.911), making it the most effective method for imputing ADS-B/GPS data.

Using real-world flight data, the authors of [72] proposed a non-invasive trust evaluation system to identify and mitigate attacks on ADS-B-based air traffic surveillance. The developed system effectively identified ADS-B spoofing, GPS spoofing, and Sybil attacks in airspace under the observation of three benign sensors, proving the efficiency of crowdsourced data in improving ADS-B security. Given the increasing usage of ADS-B protocol in uncrewed aerial vehicles (UAVS), this study concentrates on message injection attack detection and mitigation, where modified and false ADS-B messages are introduced in the airspace to mislead UAVS and ATC [73]. The C-SVM model detected message injection attacks with an accuracy of 95.32%, indicating the potential of ML models to improve UAV navigation safety. Table 5 presents the state-of-the-art ML and DL-based solutions summary.

1) ADVERSARIAL ML RISKS:

ML models for ADS-B are also vulnerable to adversarial ML risks that differ from traditional network IDS. Attackers can create perturbations at the RF or message level, such as replay attacks, precisely timed injections, bit-flip manipulations, or adaptive spoofing that exploit detector blind spots. Additionally, poisoning attacks on shared training sources can bias models. Therefore, research should evaluate robustness using adversarial benchmarks (e.g., holdout adversary

TABLE 5. Summary of State-of-The-Art ML-Based Solutions for ADS-B [58]

Ref.	Contribution and Dataset	Models	Accuracy
[59]	For jamming attack detection, the authors proposed a framework based on the simulated dataset (the authors did not mention the details of the dataset and the number of messages used in the experiments).	LR	65.6%
		SVM	67.3%
		KNN	74.6%
		ANN	81%
		DT	74.2%
[60]	The authors proposed a multi-class classification framework for ADS-B attack detection based on the Flightradar24 dataset (the authors used a dataset containing messages of 1 flight from Lisbon to Paris).	SVM	91%
		DT	92%
		RF	90%
[61]	The authors proposed a framework for the classification of aircraft and messages based on a simulated dataset (the author did not mention the number of legitimate and nonlegitimate messages).	DNN	96.66%
[62]	The authors proposed an IDS for ADS-B based on the dataset collected from OpenSky and OpenScope (the dataset containing 26,000 messages); the shared dataset link is empty.	LR	52.10%
		NB	82.10%
		K-NN	99.57%
[63]	The authors proposed a real-time IDS based on a dataset collected from OpenSky (a dataset with 20,000 messages, 10,000 legitimate and 10,000 nonlegitimate).	SVM	80%
[64]	Using DL, the authors proposed malicious message identification based on the OpenSky dataset (the author did not mention the number of legitimate and nonlegitimate messages).	LSTM	91.59%
[65]	The authors proposed a spoofing attack detection framework based on the simulated dataset (the authors did not mention the dataset details).	LSTM	93%
[66]	For spoofed message detection, the authors proposed a framework based on the simulated dataset (18,675 messages were collected for experiments).	GAN	98.87%
		CNN	99%
[67]	For message modification attack detection, the authors proposed a framework based on the OpenSky dataset (20 randomly chosen flights were downloaded from the OpenSky Network).	LSTM	98%
[68]	The authors proposed an anomaly detection model based on the Flightradar24 dataset (the authors did not mention the details of the dataset).	LSTM with GAN	97%

strategies, transfer attacks), consider adversarial training or certified-robust methods where feasible, and assess risks from model update pipelines.

2) DATASET LIMITATIONS

ML-based IDS for ADS-B shows promising results but is currently limited by dataset constraints that reduce generalisability. Available dataset sources, such as FlightRadar24 and OpenSky, provide valuable real-world traces; however, they are often incomplete, unreliable, lack adversarial labels, and contain limited realistic attacks. Due to these constraints, several studies rely on small simulated datasets that do not reflect field variability (noise, receiver diversity, multipath, antenna patterns). To promote reproducible research and realistic assessment, IDS studies should clearly report data provenance, class balance, and how attack traces were generated or recorded, and include cross-site / cross-time validation instead of relying solely on single-site holdouts.

3) FALSE ALARMS AND OPERATIONAL IMPACT:

False alarms and operational impact are critical, often under-reported metrics; even modest false positive rates can overwhelm controllers or trigger costly mitigations. Researchers should use standard detection metrics such as TPR/FPR, precision/recall, ROC/PR curves, along with detection latency and the operational cost of false alarms. They should also

explore mitigation tactics such as ensemble scoring, confidence-based escalation, human-in-the-loop thresholds, corroboration with orthogonal sensors (MLAT/PSR/RF fingerprinting), and graduated response policies. When presenting results, authors should include sensitivity studies that demonstrate how performance varies with SNR, receiver density, and the proportion of injected data.

B. CRYPTOGRAPHY-BASED SOLUTIONS

Cryptographic solutions aim to secure ADS-B communication by ensuring data integrity and authenticity. Lightweight encryption algorithms, such as Elliptic Curve Cryptography (ECC), TESLA and hybrid cryptographic techniques, are particularly suitable for resource-constrained ADS-B systems. These techniques protect against message injection, spoofing, message tampering and unauthorized access while maintaining compatibility with existing infrastructure.

To enhance ADS-B security through a key management module, the ADS-Bsec framework [77] enables secure message authentication and integrity verification during flight transitions across different air traffic control zones whereas in [75], the author proposes a lightweight, symmetric cryptography-based authentication protocol for ADS-B systems, ensuring message integrity, confidentiality, and resistance against attacks such as injection, eavesdropping, and denial of service (DoS). The proposed framework requires

secure and synchronized communication channels for initial key distribution. A Hierarchical Identity-Based Signature (HIBS) scheme was proposed [78] for ADS-B systems that ensures message authenticity and integrity while enabling batch signature verification to reduce computational overhead. The scheme relies on bilinear pairing operations, which are computationally expensive for real-time systems.

In [79], authors develop a Denial-of-Service (DoS) immune solution for ADS-B communications using location-based message authentication and enhanced timestamp, ensuring security against forged and replayed messages in aviation systems. In contrast, authors in [74] propose a Hierarchical Anonymous Certificateless Authentication (HACA) protocol for ADS-B to enhance privacy, security, and efficiency while addressing key management challenges and certificate overhead. In [80], the authors present a comprehensive survey on cryptographic methods for securing Uncrewed Aircraft Systems (UAS) and ADS-B communication links, focusing on lightweight cryptographic approaches suitable for UAV Traffic Management (UTM) systems, which require efficient, secure, and scalable solutions. In [76], the authors propose an efficient anonymous authentication scheme with batch verification for ADS-B systems, ensuring data integrity, preserving privacy, and authenticating sources while reducing computational and communication overhead.

In [81], the authors introduce a blockchain-integrated certificateless signature scheme (ECB-CLS) to improve the security of the ADS-B system. The scheme utilises elliptic curve cryptography (ECC) for lightweight, batch signature verification and employs blockchain for decentralised key management and traceability. To address packet loss, Reed-Solomon coding is incorporated for resilience. Experiments indicate ECB-CLS enhances computational efficiency, reduces signature size, and increases robustness while adding minimal overhead, making it suitable for resource-constrained ADS-B environments. The authors in [82] present a blockchain-based distributed management architecture to tackle security and performance limitations of the ADS-B system. The proposed solution incorporates CLSS for lightweight identity verification and introduces a Verifiable Random Function-based Byzantine Fault Tolerance (VRBFT) protocol with threshold signatures to enhance transaction efficiency. The results show low authentication delay (<30 ms), rapid block generation (<200 ms), and high throughput, demonstrating its practicality for real-time, large-scale intelligent vehicle management.

This work [83] proposes SAT, a backwards-compatible protocol for ADS-B message authentication using a tailored TESLA scheme, optimised for air traffic requirements. It innovatively adapts satellite-based clock synchronisation and PKI hierarchies, but its experimental validation remains limited to SDR-based lab tests without in-field trials. In [84], the authors introduce a TESLA-based broadcast authentication technique for the aviation broadband communication system, addressing man-in-the-middle and forgery attacks using SM3 hashing, whereas in [85] authors proposes CABBA, a

TESLA-based approach enhanced with phase overlay modulation and PKI, achieving bandwidth efficiency, backwards compatibility, and message authentication for ADS-B. However, the study lacks large-scale deployment analysis and assumes ideal channel conditions, leaving its real-world scalability and resilience in congested or hostile environments underexplored [85]. Authors in [86] demonstrate a real-world flight trial of secure Ground-Based Augmentation System (GBAS) communications over LDACS using TESLA-based broadcast authentication. However, the study is constrained by a limited ground station setup and does not evaluate interoperability or scalability under denser airspace or international standards.

Table 6 presents the most updated cryptography-based solutions for ADS-B security.

C. BLOCKCHAIN BASED SOLUTIONS

Blockchain technology offers a tamper-proof and decentralized method for securing ADS-B communications. Blockchain ensures data integrity and transparency by recording ADS-B messages on an immutable ledger. This technique is particularly effective in preventing message injection, spoofing, and replay attacks, as it enables real-time verification of message authenticity without relying on centralized authorities.

To enhance ADS-B security by integrating Format-Preserving Encryption (FPE) using the Feistel-based FFX algorithm and a permissioned blockchain scheme [36] proposed a technique to ensure message confidentiality, integrity, and efficient key management. In contrast, authors in [93] propose a model for securing ADS-B-based Air Traffic Management (ATM) systems using the Ethereum blockchain to ensure message authenticity, confidentiality, and integrity while addressing cyber threats in modern aviation. The model introduces computational overhead due to the blockchain framework. To enhance the security of ADS-B communications and enable reliable authentication for uncrewed aircraft systems (UAS) [88] propose a blockchain-based framework and fountain codes to ensure compatibility with existing ATM protocols. The framework requires computational overhead from blockchain operations and fountain code encoding.

In [87], authors proposed a microservices-based architecture using Blockchain to secure ADS-B data, enable message authentication and validation, and enhance anomaly detection capabilities in air traffic control systems. In contrast, a blockchain-based technique called ADS-Bchain is proposed by [89] to improve the security of ADS-B systems by preventing counterfeiting attacks, ensuring data integrity, and enabling cross-domain authentication. In [82], the authors address the security and performance limitations of ADS-B by introducing a blockchain-based distributed management architecture for intelligent vehicles. It integrates certificateless short signatures (CLSS) for lightweight authentication, ensuring integrity and non-repudiation. To further improve efficiency, a Verifiable Random Function-based Byzantine Fault

TABLE 6. Summary of Recent Cryptography-Based Solutions for ADS-B

Ref.	Contribution	Algorithm	Limitations
[36]	Introduces a combined format-preserving encryption (FPE) and blockchain-based key management framework to secure ADS-B communications.	Format-Preserving Encryption (FFX)	Requires computational resources for encryption; potential latency due to distributed ledger implementation.
[74]	The authors proposed a hierarchical certificate authentication protocol for ADS-B systems, ensuring privacy and reducing verification costs.	Certificateless authentication with CLAS	Susceptible to sophisticated Key Generation Center (KGC) attacks in certain deployment scenarios.
[75]	Proposes a symmetric cryptography-based lightweight protocol for efficient and secure ADS-B message authentication.	HMAC-SHA256, AES-CMAC	Relies heavily on symmetric key distribution; challenges in scalability for large-scale deployments.
[76]	Introduces a mutual authentication scheme with batch verification for ADS-B data integrity and source authentication.	Message Recovery Signature (MRS)	Computational overhead; challenges in scaling with an increased number of aircraft.
[77]	Develops a cryptographic key management module integrated with ADS-Bsec framework for secure key exchange across FIRs.	Keyed Hash Message Authentication Code (HMAC)	Computational overhead and potential scalability issues in high-traffic aviation zones.

TABLE 7. Summary of Blockchain-Based Solutions for ADS-B Security

Ref.	Contribution	Algorithm	Limitations
[79]	Provides a blockchain-based framework for message integrity and distributed authentication for ADS-B messages.	Distributed ledger systems	Requires substantial integration effort with existing ADS-B infrastructures.
[87]	Proposed a decentralized microservices architecture using blockchain to authenticate and validate ADS-B messages.	Blockchain with Microservices	High computational overhead and latency in real-time scenarios; lacks experimental validation.
[88]	Uses blockchain for secure flight plan authentication and ADS-B protocol enhancement.	Fountain Code, Blockchain	High latency in real-time operations; limited integration with legacy aviation systems.
[89]	Designed a trusted service scheme leveraging blockchain for anticounterfeiting ADS-B messages and ensuring integrity and authenticity.	Blockchain-based Authentication	Requires integration of multiparty systems; assumes trusted initial conditions for nodes.
[90]	Proposes a blockchain-based reputation mechanism integrated with BiLSTM-MLP for UAV position estimation and decentralized authentication.	BiLSTM-MLP, Blockchain	High computational complexity; requires consistent signal strength measurements for accuracy.
[91]	Introduces blockchain for decentralized authentication of ADS-B messages to ensure integrity and traceability.	Distributed Ledger Technology	Computational overhead; scalability issues in high-traffic aviation environments.
[92]	Proposes ATMChain, an alliance blockchain for secure, decentralized, and traceable data sharing in air traffic management.	Improved DPoS, Merkel Tree	High latency in high-data-load environments; complex implementation for integration with existing ATM systems.
[93]	Proposes Ethereum blockchain to secure ADS-B-based air traffic management by ensuring message authenticity and traceability.	Ethereum Smart Contracts	High computational and latency overhead for real-time aviation scenarios.

Tolerance protocol (VRBFT) with threshold signatures is developed.

To enhance UAV security in low-altitude airspace [90] proposes a blockchain-based dynamic reputation mechanism combined with a BiLSTM-MLP model for accurate position estimation, ensuring decentralized access control and reliable ADS-B communication. To enhance the security of Air Traffic Management (ATM) systems [92], ATMChain was developed. This blockchain-based framework ensures data integrity, access control, and authentication across ATM components by utilising an enhanced Delegated Proof-of-Stake (DPoS) mechanism. The system requires high computational requirements due to blockchain operations and off-chain storage integration [92]. Table 7 presents the most recent studies on blockchain-based solutions for protecting ADS-B from cyber attacks.

1) BLOCKCHAIN LIMITATIONS AND PRACTICAL ROLES

Blockchain-based solutions for the ADS-B system are suitable because they offer tamper-proof logging and auditable provenance. However, from a practical standpoint, most public and many permissioned ledgers experience consensus and propagation delays (seconds or more) and impose significant CPU and cryptographic costs, which conflict with aviation's sub-second surveillance and tight SWaP-C constraints.

Computational, energy, and overhead costs (including consensus, signature verification, and storage) are particularly challenging for airborne equipment and large ground deployments, where throughput and retention increase with the volume of traffic. Therefore, ledger designs should be customised for functions that can tolerate slower time-care forensics, post-incident audits, and regulatory provenance, or implemented as permissioned/hybrid systems using lightweight

TABLE 8. Summary of Most Recent Fingerprinting-Based Solutions for Protecting ADS-B From Cyber Attacks

Ref.	Contribution	Algorithm	Limitations
[80]	Introduced RF fingerprinting for ADS-B signal authentication to distinguish between legitimate and spoofed signals using SVM.	SVM for fingerprint classification	High dependency on Signal-to-Noise Ratio (SNR) and limited testing on large datasets.
[17]	Proposed a fingerprinting method to identify ADS-B transmissions using Power Spectral Density (PSD) features and Support Vector Machines (SVM).	SVM with Polynomial Kernel	Limited dataset size; performance is dependent on high Signal-to-Noise Ratio (SNR).
[94]	Introduces an RF fingerprinting approach based on zero-padding preprocessing and CNN classifiers to identify aircraft with minimal labelled data.	CNN, Zero-padding	Relies on high-quality labelled data; performance drops in low signal-to-noise ratio (SNR) environments.
[95]	Develops RF fingerprinting for detecting ADS-B message spoofing by analyzing transponder signal characteristics.	RF Signal Feature Extraction, SVM	Dependent on high-quality RF signals and stable transponder features.
[96]	Proposes a hardware fingerprinting method for ADS-B transmitters to detect spoofing using flexible and transferable ML techniques.	Siamese Networks	Limited real-world deployment scenarios reliant on high-quality signal capture and preprocessing.

BFT consensus, on-chain anchors with off-chain bulk storage, batching, and hardware-accelerated cryptography on local servers rather than full onboard ledger clients. Practical mitigations include federated, narrowly focussed ground clusters, short-lived attestations or MAC-based tokens for minimal onboard processing, and delayed verification windows that preserve immutability benefits without necessitating real-time consensus.

The newly blockchain-based solutions must report quantitative evaluations, such as end-to-end append and verification latency, throughput, per-message CPU/energy consumption, storage growth, and impacts on ATC decision latency and operator workflows. In summary, hybrid ledgers can act as an audit or forensic layer, but blockchain in its typical public form is not a ready-made solution for real-time ADS-B authentication unless prototype measurements demonstrate compliance with aviation timing, energy, and certification standards.

D. FINGERPRINTING-BASED SOLUTIONS

Fingerprinting approaches leverage unique transmission characteristics, such as frequency, signal strength, and timing variations, to identify and authenticate ADS-B transmitters during communication. These security solutions benefit by differentiating legitimate aircraft from spoofed signals, adding an additional layer of security at the physical layer.

To enhance the security of ADS-B transmissions, [17] proposed a real-time Radio Frequency Fingerprinting (RFF) method for aircraft identification using ADS-B signals, employing spectral features and ML classifiers. In [95], the authors propose a method for intrusion detection in Mode S transponders using Radio Frequency (RF) fingerprinting, enabling the identification of legitimate vs. fake ADS-B messages based on unique transponder characteristics. In [96], the authors propose a hardware fingerprinting method using

Siamese networks for securing legacy aviation communication links (e.g., ADS-B, RS-485) against message manipulation and spoofing attacks. The accuracy of the proposed fingerprinting method decreases with noisy signals and increased distance from the receiver. Table 8 presents the most recent studies on fingerprinting-based solutions for protecting ADS-B from cyber attacks.

E. TIME AND LOCATION-BASED SOLUTIONS

Time and location-based approaches verify ADS-B messages by cross-referencing transmitted data with expected flight paths and timestamps. These approaches use GPS data and multilateration techniques to detect reported times or position inconsistencies and mitigate trajectory spoofing and ghost injection attacks. A reliable surveillance framework called ATBAS is proposed by [97] using an actual Time Difference of Arrival (TDoA)-based augmentation system to enhance the accuracy of aircraft localization and detect spoofing attacks in ADS-B systems without modifying the ADS-B protocol. The proposed framework requires dense sensor deployment for accurate localization, and computational complexity may increase in larger surveillance areas. The authors use TDoA and k-nearest neighbours (k-NN) techniques. Authors in [98] develop a methodology for detecting aircraft location drifting attacks in ADS-B messages by cross-verifying receiver-estimated trajectories with the transmitted location data from ADS-B messages.

The authors propose LoVe [99], a lightweight and privacy-preserving ADS-B message verification approach that utilizes geospatial indexing (H3) and distributed public sensors to evaluate the plausibility of reported aircraft locations. The technique requires a historical database of trusted sensor data for initial setup. Table 9 presents the most recent studies on time and location-based solutions for protecting ADS-B from cyber attacks.

TABLE 9. Summary of Most Recent Time and Location-Based Solutions for Protecting ADS-B

Ref.	Contribution	Algorithm	Limitations
[79]	Introduces a timestamp-based authentication system to detect and mitigate DoS attacks on ADS-B communication channels.	Enhanced timestamp mechanism	Relies on precise timestamp data, which may be vulnerable to inaccuracies under high dynamics.
[97]	Utilizes an actual TDoA-based fingerprint map to verify ADS-B positioning data and detect spoofing attacks.	k-Nearest Neighbors (k-NN)	Limited by computation time and reliance on dense sensor network geometry.
[98]	Introduces a receiver-based trajectory cross-check method to detect location-drifting attacks in ADS-B systems.	k-Nearest Neighbors (k-NN), Fréchet Distance	Limited accuracy in low-density receiver networks; dependency on receiver placement for high precision.
[99]	Proposes a lightweight geospatial indexing system for location verification using public sensors.	Geospatial Indexing, Plausibility Mapping	Limited scalability in sparse sensor networks; assumes consistent coverage by public sensors.

F. KALMAN FILTER-BASED SOLUTIONS

Kalman filter techniques predict and verify aircraft trajectories based on historical and real-time ADS-B data. By comparing observed positions with expected values, kalman filter-based solutions can detect anomalies such as sudden deviations, unrealistic flight paths, etc., and enhance the reliability of ADS-B communication.

For detecting GNSS and ADS-B cyber attacks such as jamming, spoofing, and message injection [100] propose a crowd-sensor network combined with a Two-Step Kalman Filter (EKF). The proposed technique requires synchronization among sensor nodes, which may introduce delays. It relies on signal processing and statistical validation instead of encryption or authentication.

G. SIGNAL AND PHYSICAL LAYER-BASED SOLUTIONS

Signal and physical layer-based approaches focus on securing the communication channel by leveraging signal obfuscation, frequency hopping, and beamforming approaches. These security solutions make it difficult for attackers to intercept or jam ADS-B signals, providing robust protection against physical-layer threats.

A physical-layer-based method proposed by [101] for ADS-B message verification using Direction of Arrival (DoA) and Received Signal Strength (RSS) to improve resilience against spoofing and jamming attacks. In [101], authors combine DoA and RSS measurements to validate ADS-B messages, whereas in [102], the authors propose a backwards-compatible physical layer protocol for ADS-B systems that integrates phase modulation for message authentication without increasing channel congestion or degrading system performance. Aviation and academic researchers explored the vulnerability of ADS-B protocol to replay attacks, where the captured ADS-B messages are modified and rebroadcast to mislead air traffic controllers or originate false targets [105]. Using an RTL-Software Defined Radio (SDR) receiver, a computer-based ADS-B protocol was developed to capture over 14 million live ADS-B messages. A data-driven anomaly detection algorithm using cosine similarity was applied to identify and mitigate replayed ADS-B messages.

In [103], the authors propose a time-domain ADS-B signal separation algorithm to address signal collision and overlap issues in space-based ADS-B systems, enabling accurate message decoding and reducing bit error rates. The method requires a significant power difference between overlapping signals for effective separation. In [104], authors explore and experimentally validate interference-based attacks (constructive, destructive, and delayed interference) on ADS-B signals using Software-Defined Radios (SDRs) to highlight security vulnerabilities and propose initial considerations for countermeasures. Table 10 presents the most recent studies on signal and physical layer-based solutions for protecting ADS-B from cyber attacks.

H. SPREAD SPECTRUM TECHNOLOGY

In wireless communication networks, spread spectrum technology, including frequency hopping and direct sequence spread spectrum, minimizes or stops eavesdropping and jamming [58]. However, the network entities (sender and receiver) must pre-share a frequency-hopping mode or spreading code while using this technology. It has issues similar to cryptography, including key generation, distribution, and management.

The authors in [106] discard the pre-shared spreading modes and codes problem. Unlike the previously shared spreading code, uncoordinated spread spectrum communication does not require the sender and receiver to share the spreading code prior to communication. Instead, they randomly switch between channels or employ spreading codes, making it difficult for attackers to effectively jam or eavesdrop on the channel. The drawback is the inefficient use of bandwidth resources, as communication parties (sender and receiver) are often not on the same channel. Despite the effectiveness of spread spectrum technology in countering diverse attacks, its inherent limitations, such as extended time requirements and low performance, pose challenges to its application in ADS-B systems [107].

I. COMPARATIVE OVERVIEW OF DISCUSSED SOLUTION

Table 11 provides a brief, qualitative comparison of the discussed ADS-B security solutions, focusing on

TABLE 10. Summary of Most Recent Signal and Physical Layer-Based Solutions for Protecting ADS-B From Cyber Attacks

Ref.	Contribution	Algorithm	Limitations
[101]	Proposes a physical-layer signal authentication method using DoA and RSS estimation for spoofing and jamming detection.	DoA Estimation, RSS Analysis	Requires complex antenna systems; vulnerable to signal power manipulation during RSS-based verification.
[102]	Proposes a physical-layer protocol using phase modulation for backwards-compatible ADS-B message authentication.	Binary Phase Shift Keying (BPSK), QPSK	Limited by hardware constraints in legacy systems; challenges in integrating with existing infrastructure.
[103]	Proposes a time-domain cancellation algorithm to separate overlapping ADS-B signals effectively.	Singular Value Decomposition (SVD)	Limited performance in scenarios with high noise or dense overlapping signal conditions.
[104]	Explores interference attacks on ADS-B transmissions, evaluating the effects of constructive, destructive, and delayed interference.	Matched Filters, Convolution Filters	Limited testing scenarios; results based on simulated environments without real-world validation.

TABLE 11. Comparative Summary of ADS-B Existing Solutions

Solution class	Computation	Latency	Scalability	Deployment cost	Detection accuracy	Typical timeframe	False-positive risk
Machine Learning and Deep Learning	High	Low	High	Medium	Medium-High (data dependent)	Immediate-Short	Medium-High (data bias/tuning)
Cryptography-Based Solutions	High	Medium-High (per-frame verify)	Medium	High	Very High (integrity)	Long	Low
Blockchain-Based Solutions	High	High	Low-Medium (design dependent)	High	High	Medium-Long	Low
Fingerprinting-Based Solutions	Medium	Low	Medium	Medium	Medium-High	Short-Medium	Medium
Time and Location-Based Solutions	Low-Medium	Low	Medium	Medium-High (sensors)	Medium-High	Immediate-Short	Medium
Kalman Filter-Based Solutions	Low	Low	High	Low-Medium	Medium	Immediate	Medium
Signal and Physical Layer Solutions	Medium	Low	Medium	Medium-High (antenna/hardware cost)	High (jamming/spoof detection)	Short-Medium	Low-Medium
Spread Spectrum Technology	Medium	Low-Medium	Medium	High (airborne changes)	High	Long	Low

computation, latency, scalability, deployment cost, detection accuracy, false-positive risk, and realistic timeframe. Ground-side approaches, including ML and DL, Kalman filter, MLAT/DoA, and fingerprinting, generally provide low latency, manageable computation (for inference or light filtering), high scalability, and quick to short-term deployment. However, their accuracy and false-positive rates rely on data quality and sensor coverage. Cryptographic and spread-spectrum approaches offer high integrity and anti-jam advantages, but they require significant computation, retrofit costs, and lengthy certification processes. Blockchain ensures authentication and integrity but suffers from high commit latency and ground-side expenses, making it more suitable for audit and logging rather than real-time authentication. In practice, the suggested near-term strategy is layered: first, deploy ground-side, low-latency defenses for quick gains; then, use sensor fusion for corroboration; and finally, reserve resource-intensive cryptographic or protocol modifications for long-term, regulator-backed deployments.

VIII. TAXONOMY: INTEGRATED VIEW OF ADS-B SECURITY

The landscape of ADS-B security is multi-dimensional and complex, shaped by technological constraints, vulnerabilities, and the critical need for aviation safety. This section extracts key findings from our survey paper, aligning them with the integrated taxonomy (Fig. 6) to provide a holistic understanding of current solutions, threats, limitations, and future prospects.

A. THREAT LANDSCAPE AND SECURITY GOALS

ADS-B protocol is an open broadcast architecture that makes it vulnerable to a wide range of threats such as message flooding, deletion, modification, injection, eavesdropping, and jamming. These attacks directly compromise air traffic surveillance systems’ core cybersecurity goals: Confidentiality, Integrity, Authentication, and Availability (CIAA). Most notably, the lack of mutual authentication and encryption techniques makes spoofing and ghost aircraft injection feasible and scalable for attackers with low-cost SDRs. The existing

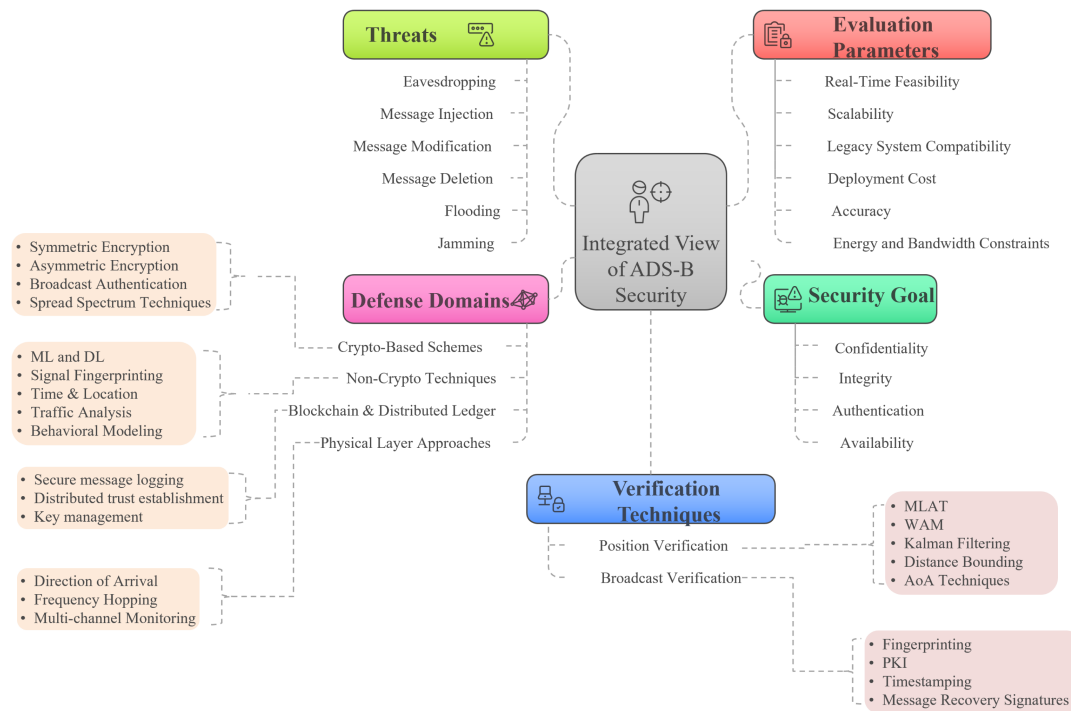


FIGURE 6. Taxonomy: Integrated View of ADS-B Security.

research consistently features these vulnerabilities, affirming that no single solution adequately addresses all facets of ADS-B insecurity.

B. OPERATIONAL CONTEXT AND FEASIBILITY

The security solutions proposed for ADS-B must be technically sound and operationally feasible. For example, ML-based IDS require inference times below 200 ms to qualify as real-time. Likewise, public blockchains introduce latency in the order of seconds, which is unacceptable in high-density airspaces. The suitability of any proposed defence technique is thus deeply tied to these aviation-specific constraints. Besides this, the role of technologies like TCAS should be clarified. TCAS is designed for airborne collision avoidance, whereas ML and DL detection models are better suited for ground-based threat monitoring or distributed onboard inference.

C. VERIFICATION TECHNIQUES AND EVALUATION METRICS

Verification mechanisms, including broadcast verification (e.g., TESLA, fingerprinting) and position verification (e.g., MLAT, Kalman filtering, Distance Bounding), form the backbone of ADS-B security architectures. Each has merits and trade-offs: Fingerprinting and timestamping excel in detecting subtle tampering but are sensitive to noise and require extensive signal analysis. At the same time, Kalman filters are lightweight but less effective against sophisticated spoofing. When assessing these security approaches, we must consider critical operational parameters, such as accuracy, deployment cost, real-time feasibility, bandwidth constraints and compatibility with legacy systems. For example, ICAO standards

dictate that position updates must occur at intervals of less than 250 milliseconds; hence, cryptographic solutions with processing delays exceeding this threshold are impractical.

D. DEFENCE DOMAINS: A MULTIDISCIPLINARY ARSENAL

The defence ecosystem can be broadly categorised into four main domains: cryptographic, non-cryptographic, physical-layer and blockchain-based frameworks.

- *Cryptographic-based solutions* such as HMAC and TESLA show strong promise in message authentication and integrity. However, their deployment in aviation is hindered by limited processing power and real-time constraints on legacy avionics.
- *Non-crypto approaches* such as ML and DL, time/location and fingerprinting focus on anomaly detection and verification without modifying the core structure of the ADS-B. Non-crypto approaches are valuable for ground-based systems but raise issues of model drift, training data scarcity, and deployment complexity in airspace environments.
- *Physical-layer solutions* such as Direction of Arrival (DoA) estimation and frequency hopping add robustness against spoofing and jamming but often require hardware upgrades, making large-scale deployment impractical in the short term.
- *Blockchain-based solutions* offer decentralised message authentication and integrity but experience high latency and computational overhead. Permissioned blockchain architectures may provide a middle ground, balancing decentralisation with performance feasibility.

E. TOWARDS AN INTEGRATED DEFENCE FRAMEWORK

ADS-B attacks operate across both the Physical and Data-Link layers and ground/backhaul domains, targeting different security properties of CIA+A. Because aviation places hard constraints on latency, airborne SWaP-C, and certification timelines, no single countermeasure can meet all requirements. For these reasons, we propose a pragmatic, layered defence-in-depth strategy, including fast and low-cost airlink checks to detect obvious faults; corroborative multi-sensor checks to reduce false alarms; and hardened backend analytics and procedures to perform final decision-making and forensic logging. Considering the diversity of ADS-B threat vectors and the limitations of available solutions, the security of ADS-B must be addressed through a multi-layered defence-in-depth approach. The multi-layered defence-in-depth involves:

- Improving physical-layer security through directional antennas and spectrum-spreading techniques.
- Deploying ML-based detection models on ground control systems.
- Using lightweight cryptographic authentication for message integrity.
- Leveraging permissioned blockchain for distributed logging and consensus-based authentication.

A hybrid framework can significantly enhance ADS-B resilience while supporting global interoperability standards by striking a balance between technical robustness and aviation-specific feasibility.

IX. CHALLENGES, FUTURE AND CROSS-CUTTING CHALLENGES

A. OPEN CHALLENGES IN ADS-B SECURITY

Developing efficient and robust security methodologies for ADS-B protection involved several technical, regulatory, and operational challenges. Based on insights from the existing literature, below is the outline of key challenges in developing efficient and effective ADS-B security solutions.

- *Computational and Latency Constraints:* ADS-B operates in real-time and requires instantaneous data transmission and processing. Developing and implementing anomaly detection and cryptographic authentication can introduce unacceptable delays for such critical aviation safety systems. The major challenge is to design and develop lightweight security solutions that have low latency in data transmission, are computationally efficient, and are compatible with existing avionics.
- *Compatibility with Legacy Aviation Systems:* Most commercial aircraft worldwide have already deployed ADS-B for location transmission; such large-scale modifications make it impractical to modify the protocol [25]. The developed security solutions must be backward-compatible with already deployed ADS-B infrastructure, which limits the feasibility of most of the proposed solutions.
- *Cost and Deployment Challenges:* Retrofitting existing aircraft with hardware-based security solutions can

be logistically complex and expensive. Ground-based anomaly detection systems require limiting widespread adoption and substantial infrastructure investment. Airlines and air traffic controllers may be unwilling to invest in security solutions without immediate financial benefit.

- *False Alarms in Existing Security Solutions:* The existing research explores AI-based intrusion detection systems, including ML and DL for ADS-B security. However, the proposed solutions don't provide comprehensive protection against the identified attacks. They usually struggle with high false positive rates, triggering unnecessary false alarms and potentially increasing the workload for ATCs. Further research is required to develop robust, efficient, fine-tuned AI models for ADS-B anomaly detection with low false positives.
- *Coordinated and Sophisticated Cyber Attacks:* Most current research addresses isolated attack vectors, and comprehensive, multi-layered attack scenarios remain under-explored. Sophisticated threat actors could target ADS-B systems as part of coordinated cyber-physical attacks, potentially combined with GNSS spoofing and communication disruptions.
- *Dataset Limitations and Availability:* Datasets used in ADS-B research for IDS development are helpful but limited to public feeds, such as OpenSky, FlightAware, FlightRadar24, ADS-B Exchange, and FAA, and lack adversarial labels, types of attacks, and often have access restrictions. Additionally, many studies depend on small or simulated datasets (OpenScope) that fail to reflect real-world noise, receiver diversity, or attack realism. This leads to poor generalisability and a high risk of overfitting; therefore, researchers should report provenance, class balance, and cross-site validation. Publishers should prioritise shared, labelled, multi-site datasets (including realistic IQ captures and attack traces) to improve benchmarking and robustness.
- *Regulatory and Standardization Challenges:* Adopting a new security methodology requires endorsement from global regulatory bodies such as the ICAO, FAA, and EUROCONTROL. These global regulatory bodies must balance security enhancements with operational practicality. However, to this day, no globally accepted ADS-B security technique exists, leading to fragmented acceptance across different regions. Hence, policy development is slow, and adoption requires alignment between governments, airlines, manufacturers, and regulators.

B. FUTURE RESEARCH OPPORTUNITIES

There is an urgent need for research and innovation in ADS-B to strengthen its resilience against evolving threats. We outline key future research directions to protect the ADS-B and foster more robust, scalable, and adaptive solutions to safeguard global air traffic systems.

- *Lightweight Cryptographic Protocols for Aviation:* The lack of authentication and encryption makes ADS-B

vulnerable to several security threats and poses a significant challenge to its security. Due to high computational overhead, traditional cryptographic techniques, such as elliptic curve cryptography and public key infrastructure, are impractical for real-time communication in applications like the aviation industry. This gap highlights a key future direction focusing on lightweight cryptography protocols compatible with ADS-B, and work on quantum-safe cryptography to protect and prepare the ADS-B protocol from future quantum computing threats [108].

- *Advanced ML-Based IDS*: As identified in Section VII, ML and DL play an important role in the security of ADS-B anomaly detection. However, the high false positive rates in the existing models limit their implementation in the real world. Future research should focus on transfer learning techniques to adapt the models to evolving attack patterns, increase the models' explainability and interpretability in attack detection, and explore the role of federated learning in ADS-B attack detection, where different ATC centres can collaborate [109].
- *Federated Learning for Distributed IDS*: Using federated learning, researchers can train anomaly-detection models across multiple ground sites without centralising raw ADS-B/IQ data, thereby enhancing detection generalisation while maintaining privacy. To evaluate detection accuracy, communication overhead, aggregation latency, and robustness, researchers need to use simulated or real multi-site splits.
- *Countermeasures for Jamming and GPS Spoofing*: ADS-B highly depends on GPS data, making it vulnerable to jamming and spoofing attacks. These attacks can disrupt surveillance and navigation, and current systems lack robust countermeasures. In future studies, researchers should focus on implementing multi-sensor data fusion technology to validate ADS-B data from different inputs for detecting anomalies in aircraft positioning [101]. Moreover, adaptive transmission schemes to minimize jamming impact should be explored.
- *Hybrid Surveillance Systems*: Relying on one surveillance system is risky. Hybrid systems offer redundancy and better security. This gap highlights that researchers should explore integrating ADS-B with other surveillance systems, such as primary/secondary radar, multilateration, and satellite-based tracking, to enhance reliability. New algorithms that combine data from multiple sources to cross-validate aircraft positions are required.
- *Real-World Validation of ADS-B Security Solutions*: As identified in Section VII, the existing proposed security solutions are mostly theoretical or limited to simulations and are not validated in real-world scenarios [110]. Future research should focus on collaboration with the ATC and airlines to deploy the developed security solutions, validate their effectiveness, and conduct large-scale

real-world experiments using ADS-B data from diverse sources.

- *Secure ADS-B Communication in Uncrewed Aerial Vehicles (UAVs)*: The continuous increase in the use of drones and UAVs in military and commercial applications increases the demand for securing ADS-B communication [111]. This gap highlights that future research should explore encrypted ADS-B transmission to secure multi-UAV coordination strategies, explore privacy-preserving protocols for UAVs and develop enhanced autonomous attack detection systems for UAV ATM [111].
- *Human Factors in ADS-B Security*: ATCs and pilots work in high-stress and time-critical environments. Introducing new security procedures or tools without considering human limitations can increase cognitive load, slow reaction times, or lead to errors in judgment. This gap highlights that future research on cognitive load when responding to security alerts and designing clear and actionable alert messaging that supports rapid decision-making without ambiguity is required.
- *Regulatory and Standardization Challenges*: As identified in Section IV-A, ADS-B security issues are globally recognised, and regulatory frameworks differ across countries, slowing interest in implementing security solutions to protect the protocol from identified threats and attacks. Future research should focus on developing universal security standards in alliance with the FAA, ICAO, and EUROCONTROL, analysing the cost-benefit tradeoffs of the available security solutions, and investigating legal and ethical considerations [112].

C. CROSS-CUTTING CHALLENGES

As the aviation industry seeks to enhance the security of the ADS-B protocol, several overarching challenges intersect with proposed solutions. These overarching challenges reflect the trade-offs between security, compatibility, performance, and scalability, which must be carefully considered during the implementation process.

- *Centralisation vs. Latency in Blockchain-Based Solutions*: Blockchain-based solutions offer decentralised, tamper-resistant mechanisms for data authentication and logging. However, block propagation and consensus algorithms significantly enhanced latency, making real-time ADS-B message validation difficult. Public blockchains, in particular, face scalability and throughput limitations. While lightweight blockchain reduces these issues, its integration into highly centralised aviation systems remains complex.
- *Compatibility with Legacy Systems*: Most of the currently used aircraft systems were not designed with modern cybersecurity in mind. Introducing advanced security mechanisms like multilateration cross-verification, cryptographic modules, or blockchain nodes may require

hardware retrofitting or firmware updates. Such modifications can be costly and operationally disruptive. Furthermore, regulatory approval and standardisation are necessary before widespread adoption, adding another layer of complexity.

- **Machine Learning Accuracy vs. Real-Time Feasibility:** ML and DL-based models have shown advancement in detecting advanced ADS-B threats such as jamming, spoofing, and injection. However, many high-accuracy models, especially deep neural networks, require significant memory and processing time, which creates challenges for real-time deployment in constrained airborne environments. Additionally, the lack of large, comprehensive attack types and labelled datasets impairs the generalisation and robustness of these models.
- **Lightweight Protocols vs. Strong Encryption:** Encryption approaches like AES, RSA, and ECC provide critical confidentiality and integrity guarantees for ADS-B. However, they increase substantial computational and memory overhead, which can exceed the capabilities of legacy avionics and embedded systems. Conversely, format-preserving encryption methods and lightweight cryptographic protocols are more resource-efficient but may offer reduced security margins. Striking the right balance is crucial to ensuring both safety and system responsiveness.

X. CONCLUSION

ADS-B protocol has significantly enhanced air traffic surveillance, efficiency, cost-effectiveness, and situational awareness. However, a lack of authentication and encryption makes it vulnerable to several security threats and attacks, posing serious risks to aviation safety. This survey critically evaluates ADS-B security and the evolving threat landscape, systematically analyzing existing vulnerabilities and the effectiveness of various mitigation methodologies. Through a comparative analysis of existing security approaches, this study highlights their feasibility challenges, real-time implementation trade-offs, strengths, and weaknesses. The proposed research survey also identifies key security challenges that hinder developing and deploying efficient and robust ADS-B security solutions. Furthermore, the research highlights the urgent need for a lightweight multi-layered security system incorporating cryptography, AI-based anomaly detection, and a real-time verification approach to enhance ADS-B resilience. As the future aviation industry advances, securing ADS-B communication remains a top priority for aviation authorities worldwide.

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